

Test Case Name :Equivalence class Analysis for triangle problem

Test Data : Enter the 3 Integer Value(a , b And c)

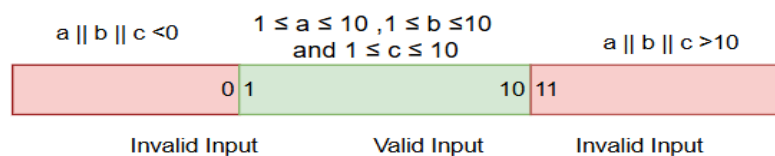
Pre-condition : $1 \leq a \leq 10$, $1 \leq b \leq 10$ and $1 \leq c \leq 10$ and $a < b + c$, $b < a + c$ and $c < a + b$

Brief Description : Check whether given value for a Equilateral, Isosceles , Scalene triangle or can't form a triangle

Source Code:

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int a,b,c;
7
8     cout<<"Enter value for side A: ";
9     cin>>a;
10    cout<<"Enter value for side B: ";
11    cin>>b;
12    cout<<"Enter value for side C: ";
13    cin>>c;
14
15    if(a<=0 || b<=0 || c<=0)
16    {
17        cout<<"Invalid input: "<<endl;
18        return 0;
19    }
20
21    if(a>10 || b>10 || c>10)
22    {
23        cout<<"Invalid Input: "<<endl;
24        return 0;
25    }
26
27    // Check for a valid triangle using the triangle inequality theorem
28    if((a+b+c) && (b<a+c) && (c<a+b))
29    {
30        if(a==b && b==c)
31        {
32            cout<<"Equilateral triangle" <<endl;
33        }
34        else if(a==b || a==c || b==c)
35        {
36            cout<<"Isosceles triangle" <<endl;
37        }
38        else
39        {
40            cout<<"Scalene triangle" <<endl;
41        }
42    }
43    else
44    {
45        cout<<"Triangle cannot be formed" <<endl;
46    }
47
48    return 0;
49 }
```

EQUIVALENCE CLASS PARTITIONING:



TEST SUMMARY:

Project Name:	Triangle Problem Decision (Equivalence Partitioning)
Test ID:	Solve_Triangle_EP_TES61
Test Title:	Use Equivalence partitioning approach for triangle problem
Test Priority:	Low
Module Name:	Identify_Triangle
Test Data:	Enter the 3 Integer Value (a, b and c)

Designed By:	Urwa Qadir
Designed Date:	2/16/2025
Executed By:	Fatima Jaffar
Executed Date:	2/16/2025
Description of Test:	Check whether the given values form an equilateral, isosceles, scalene triangle or no triangle using equivalence
Pre-Conditions:	Range in which testing is to be done: $1 \leq a \leq 10$, $1 \leq b \leq 10$ and $1 \leq c \leq 10$ and $a < b + c$, $b < a + c$ and $c < a + b$
Dependencies:	No dependencies in this test.

TEST CASES:

Test Case	Test Description	Valid			Expected Output	Actual Output	Status
		A	B	C			
TC_01	Side A is less than 1, which is out of range.	0	5	5	invalid Input	invalid Input	pass
TC_02	Side A is greater than 10, which is out of range.	11	5	5	invalid Input	invalid Input	pass
TC_03	All 3 sides are equal	5	5	5	Equilateral	Equilateral	pass
TC_04	Side B is less than 1, which is out of range.	5	0	5	Invalid Input	Invalid Input	pass
TC_05	Side B is greater than 10, which is out of range.	5	11	5	Invalid Input	Invalid Input	pass
TC_06	Side C is less than 1, which is out of range.	5	5	0	invalid Input	invalid Input	pass
TC_07	Side C is greater than 10, which is out of range.	5	5	11	invalid Input	invalid Input	pass
TC_08	Two sides (A and B) are equal.	5	5	6	Isosceles	Isosceles	pass
TC_09	Two sides (A and C) are equal.	5	6	5	Isosceles	Isosceles	pass
TC_10	Two sides (B and C) are equal.	6	5	5	Isosceles	Isosceles	pass
TC_11	All three sides are different.	10	9	5	scalene	scalene	pass
TC_12	The sum of two sides (B+C=10) is equal to A, violating the triangle inequality theorem.	10	5	5	Not a triangle	Not a triangle	pass

TC_13	The sum of two sides (B+C=8) is less than A, violating the triangle inequality theorem.	10	5	3	Not a triangle	Not a triangle	pass
TC_14	The sum of two sides (A+C=10) is equal to B, violating the triangle inequality theorem.	5	10	5	Not a triangle	Not a triangle	pass
TC_15	The sum of two sides (A+C=8) is less than B, violating the triangle inequality theorem.	5	10	3	Not a triangle	Not a triangle	pass
TC_16	The sum of two sides (A+B=8) is less than C, violating the triangle inequality theorem.	3	5	10	Not a triangle	Not a triangle	pass